

A Matrix Gain for Extremum Seeking

Zap stochastic approximation on a wind turbine: what each object becomes, and whether it works

Aykut C. Satici

Robot Control Lab · Systems Engineering · UT Dallas

Two answers, and only one needs a Hessian

Stochastic approximation offers two routes to the same asymptotic covariance. They differ entirely in what they cost to build.

Polyak-Ruppert averaging

Average the iterates. Attains the optimal asymptotic covariance with no curvature estimate anywhere.

Zap stochastic approximation

Estimate the Jacobian on a fast timescale, invert it, use it as a matrix gain. A stochastic Newton-Raphson method.

Both reach the optimal covariance $\Sigma^{PR} = A^{-1}\Sigma_W A^{-\top}$. Averaging gets there for four lines of code and no curvature estimate at all. **So if the case for a matrix gain is argued on terminal variance, it loses.** What it can be argued on is transient behavior and conditioning, and this deck asks whether that survives contact with a turbine.

Zap in four lines

$$\begin{aligned}\hat{A}_{n+1} &= \hat{A}_n + \varepsilon_{n+1} [A_{n+1} - \hat{A}_n] \\ \theta_{n+1} &= \theta_n + \alpha_{n+1} G_{n+1} f_{n+1}(\theta_n) \\ A_{n+1} &= \partial_{\theta} f_{n+1}(\theta_n) \\ G_{n+1} &= -[\varepsilon I + \hat{A}_{n+1}^{\top} \hat{A}_{n+1}]^{-1} \hat{A}_{n+1}\end{aligned}$$

with the two step sizes satisfying $\varepsilon_n / \alpha_n \rightarrow \infty$.

Line three is the whole problem. A is the Jacobian of the **sample** update direction, differentiated in the parameter. In Q-learning f_{n+1} is piecewise linear in θ , so that derivative is free and rank one. A demodulated gradient estimate has no such closed form.

The dictionary

θ_n the parameter

torque gain $u = k/l$, or (u, β) ,
or a vector of yaw angles γ

$f_{n+1}(\theta_n)$ the noisy update direction

one demodulated gradient estimate
of $\ln P$, per dither period

A_{n+1} its Jacobian in θ

the curvature of $\ln C_p$, read off the
SECOND harmonic of the same record

$\hat{A}_n \rightarrow A(\theta)$ the averaged Jacobian

the Hessian at the operating point,
which erosion moves over months

Everything Zap needs is a function of $\ln P(t)$ and the probe. No new sensor, no extra excitation.

What f is, precisely

The objective is the turbulence-averaged log-power, $\Gamma(\theta) = \mathbb{E}[\ln P(\theta)]$, and because $3 \ln V$ does not depend on θ this equals $\mathbb{E}[\ln C_P]$ up to a constant.

The mean field is $\bar{f}(\theta) = \nabla \Gamma(\theta)$. The sample is one dither period of demodulation:

$$f_{n+1}(\theta_n) = \hat{g}_n = \frac{2}{aT} \int_{nT}^{(n+1)T} \ln P(t) \sin \omega t \, dt$$

So f is exactly what LP-ESC already computes. Nothing changes in the forward path. Zap only changes what multiplies $\hat{g}_n$ before it reaches the integrator.

What A is, precisely

$$A(\theta) = \partial_{\theta} \bar{f}(\theta) = \nabla^2 \Gamma(\theta)$$

For the torque-gain problem this is the scalar $\partial^2 \ln C_P / \partial u^2$ at the operating point. It is the **same number** that appears in Rotea's 2017 crossover frequency,

$$\nu_c = \frac{\kappa}{C_P^{\max}} \left| \frac{\partial^2 C_P}{\partial u^2}(u_o) \right|$$

The curvature is not a new quantity being introduced. It is already in the tuning formula, taken on faith from a design curve. Zap proposes to measure it instead.

Where A_{n+1} comes from

The probe already writes the curvature into the data. Nobody reads it.

Expand around the operating point

$\theta(t) = \theta_n + a \sin \omega t$ is the **ordinary ESC dither**: θ_n the current estimate, held across the period, with the probe riding on it. It is $d_1(t) = a \sin(\omega t)$ of [KR22] §2, nothing added. Put it into a local quadratic model of $y = \ln P$, with g the slope and H the curvature at θ_n :

$$y(t) = \text{const} + g a \sin \omega t + \frac{1}{2} H a^2 \sin^2 \omega t$$

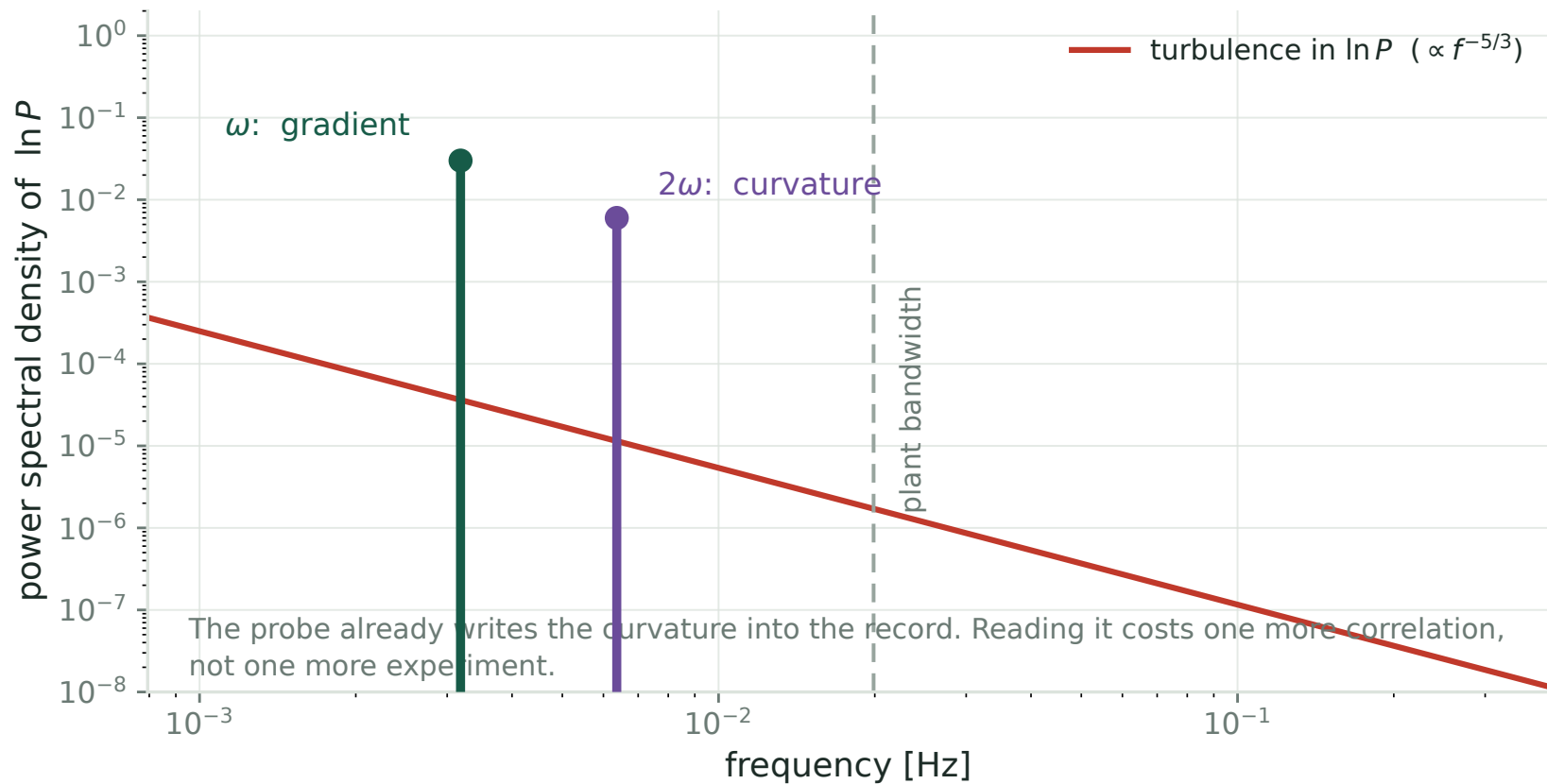
Now use $\sin^2 \omega t = \frac{1}{2}(1 - \cos 2\omega t)$:

$$y(t) = \underbrace{\left[\cdot + \frac{1}{4} H a^2 \right]}_{\text{DC}} + \underbrace{g a \sin \omega t}_{\text{first harmonic}} - \underbrace{\frac{1}{4} H a^2 \cos 2\omega t}_{\text{second harmonic}}$$

The gradient sits at ω . The curvature sits at 2ω . Same record, separated in frequency, one probe generating both.

This needs the dither to be a **sinusoid**. For the square wave of [CLR19], $g(t) = \pm 1$ so $g(t)^2 \equiv 1$: the quadratic term is a constant and **carries no harmonic at all**. The method applies to the [KR22] line, which dithers sinusoidally, and not to the 2019 square-wave line.

In the spectrum



The two demodulators

$$\hat{g}_n = \left\langle y \cdot \frac{2}{a} \sin \omega t \right\rangle, \quad \hat{H}_n = \left\langle y \cdot N(t) \right\rangle, \quad N(t) = \frac{16}{a^2} \left(\sin^2 \omega t - \frac{1}{2} \right)$$

Check the second: $N(t) = -\frac{8}{a^2} \cos 2\omega t$, averaged against the $-\frac{1}{4} H a^2 \cos 2\omega t$ term of the expansion, gives $\frac{1}{4} H a^2 \cdot \frac{8}{a^2} \cdot \frac{1}{2} = H$. For p parameters the same construction is a matrix, Eqs. 20–21 of [GKN12]:

$$N_{i,i}(t) = \frac{16}{a_i^2} \left(\sin^2 \omega_i t - \frac{1}{2} \right), \quad N_{i,j}(t) = \frac{4}{a_i a_j} \sin \omega_i t \sin \omega_j t \quad (i \neq j)$$

$A_{n+1} = \hat{H}_n$. One extra correlation against a signal the loop already generates: no new sensor, no extra excitation, no additional load.

Two conditions, both checked later by simulation. The dither must be a **sinusoid**, since a square wave has $g(t)^2 \equiv 1$ and no second harmonic at all. And 2ω must sit inside the rotor's passband, or $\hat{H}$ reads low: 21% low at the published $T = 150$ s.

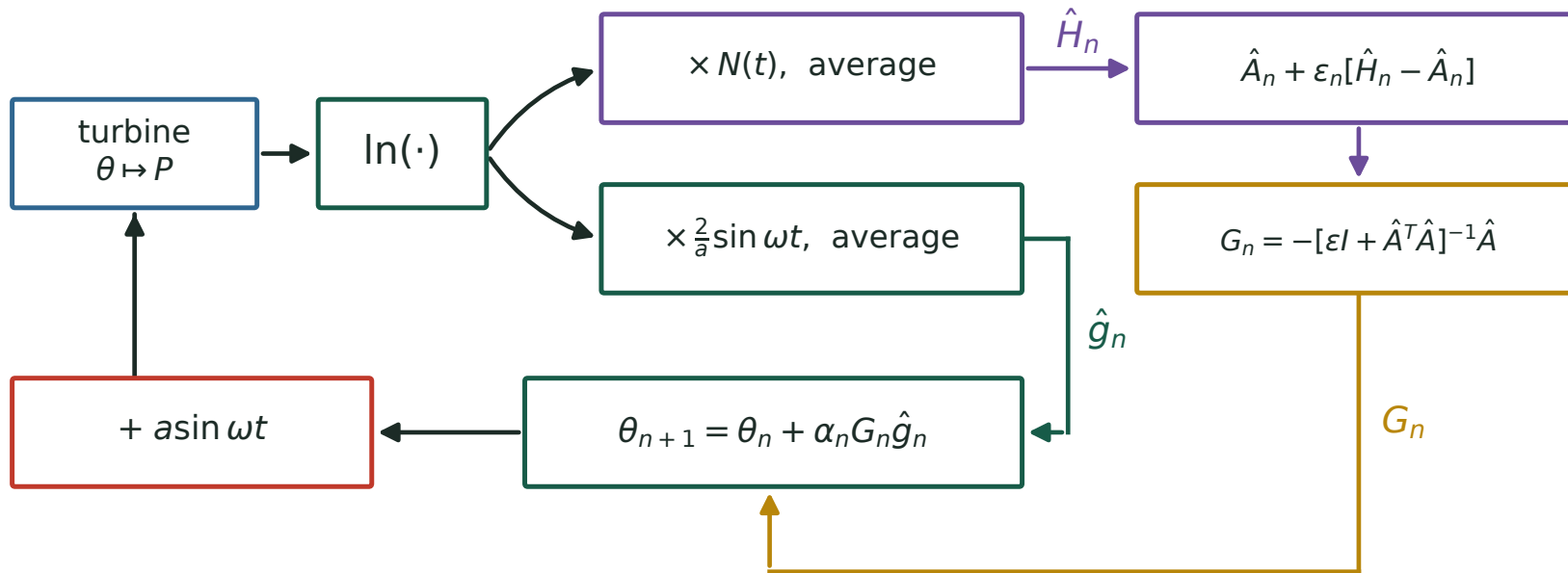
More than one parameter

For $\theta \in \mathbb{R}^p$, dither each component at its own frequency and demodulate with a matrix:

$$N_{ii}(t) = \frac{16}{a_i^2} \left(\sin^2 \omega_i t - \frac{1}{2} \right), \quad N_{ij}(t) = \frac{4}{a_i a_j} \sin \omega_i t \sin \omega_j t \quad (i \neq j)$$

The frequencies must satisfy the usual separation conditions, and now also $2\omega_i$ must not collide with any ω_j or $\omega_j \pm \omega_k$. For six yawed turbines that is a real combinatorial constraint on a narrow usable band.

The loop, assembled



$$N(t) = \frac{16}{a^2} \left(\sin^2 \omega t - \frac{1}{2} \right)$$

The lower path is the existing algorithm. Everything in violet and gold is added.

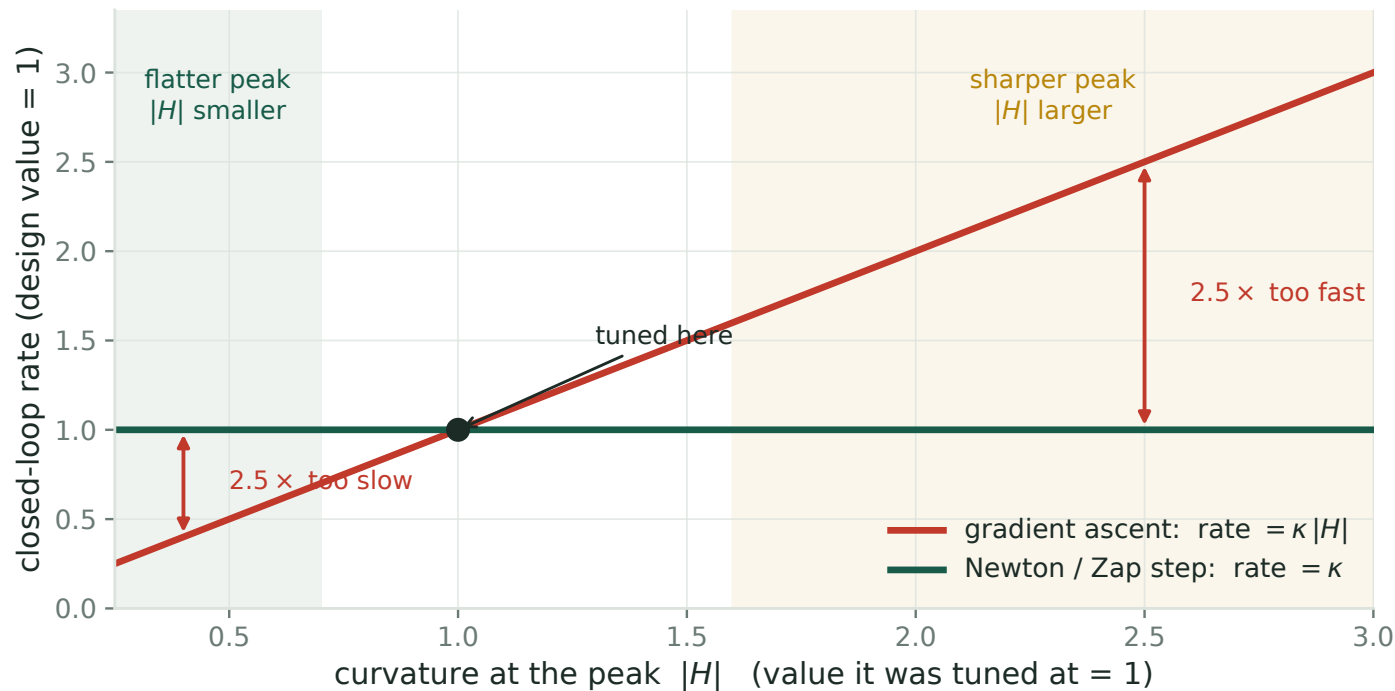
The lower path is LP-ESC exactly as published. Everything added is a second correlation and two recursions, all on signals already in hand.

What it buys

Three things, and only one of them is about noise.

The curvature leaves the loop gain

Write $\delta = \theta - \theta^*$, so $g = H\delta$ with $H < 0$ at a maximum. Gradient ascent linearizes to $\dot{\delta} = \kappa H\delta$: **the closed-loop rate is $\kappa|H|$, so it inherits the plant's curvature**. Meyn's gain carries its own sign, $G = -H/(\varepsilon + H^2)$, and with $\varepsilon \rightarrow 0$ the update gives $\dot{\delta} = -\kappa\delta$, rate κ regardless of H .



Read the diagonal: tune at $|H| = 1$, and if erosion or an off-design site flattens the peak to $|H| = 0.4$ the loop runs $2.5\times$ **too slow**; sharpen it to 2.5 and it runs $2.5\times$ **too fast**, toward the stability edge. Nothing in the loop reports either. The flat line is what a Newton step buys: κ is the settling time rather than merely influencing it.

Which is the tuning complaint, answered

What Rotea wrote

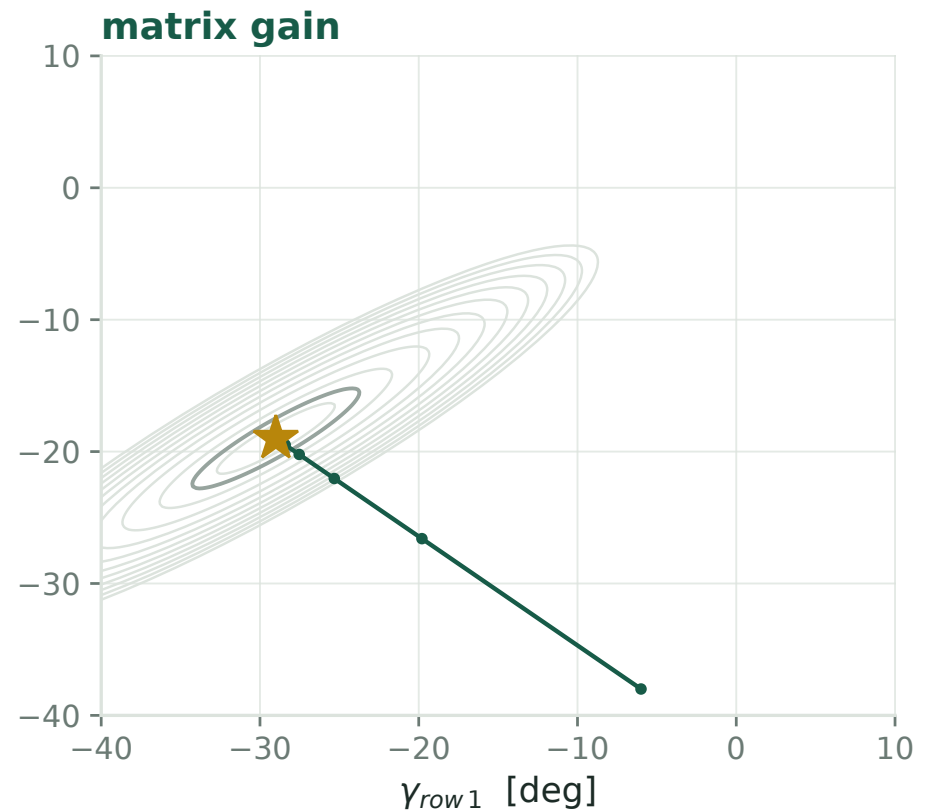
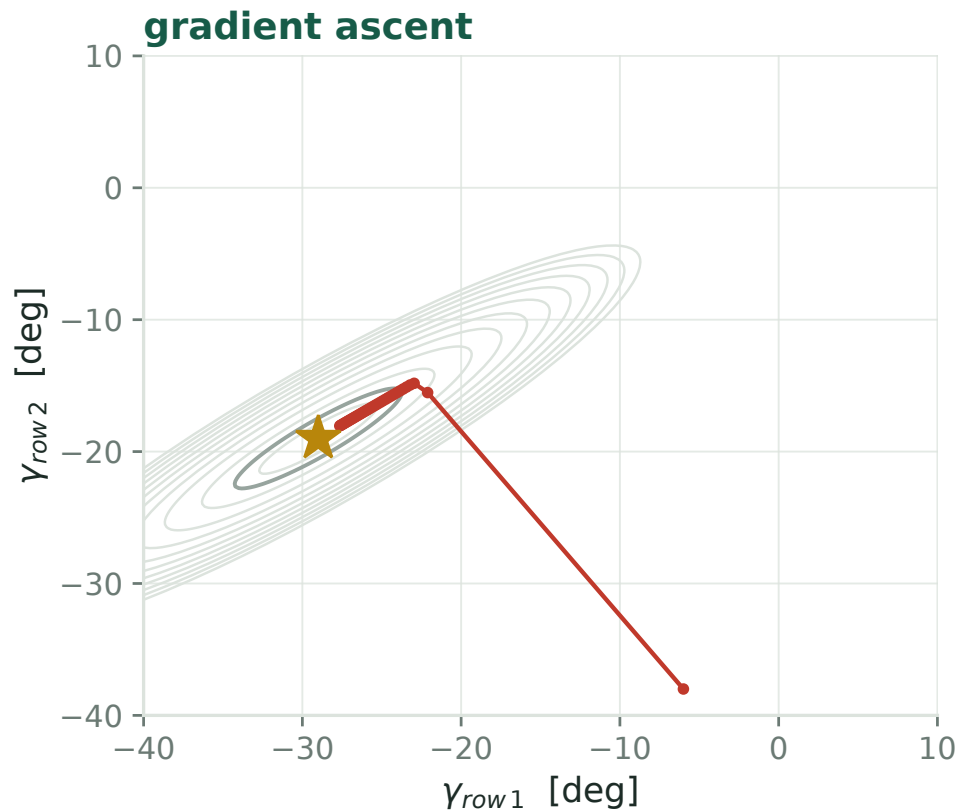
A version of ESC that “converges very fast but it takes too long to tune its parameters.”

What [RKAJ24] says

“All parameters with the exception of dither frequency and amplitude were obtained by trial and error.”

Under gradient ascent, κ has to be re-found whenever H changes, and H changes with the site, with the blade condition, and with the operating point. Under a matrix gain, κ is set once from the settling time you want.

Where a matrix really pays



Gradient methods converge at a rate set by the **smallest** Hessian eigenvalue. The measured farm power maps in the twelve-turbine tunnel experiment are long narrow ridges in the yaw angles, which is precisely the case where that hurts.

Regularization is doing real work

In the scalar case $G = -\hat{A}/(\varepsilon + \hat{A}^2)$, so the effective rate is

$$\kappa \frac{H^2}{\varepsilon + H^2}$$

On a flat peak $\hat{A} \rightarrow 0$ and an unregularized Newton step would explode. This form does the opposite: it backs off to a small step, degrading gracefully into gradient ascent exactly where curvature cannot be seen.

What it costs

Three objections. One of them is a hard gate.

Why the curvature is better conditioned near the peak

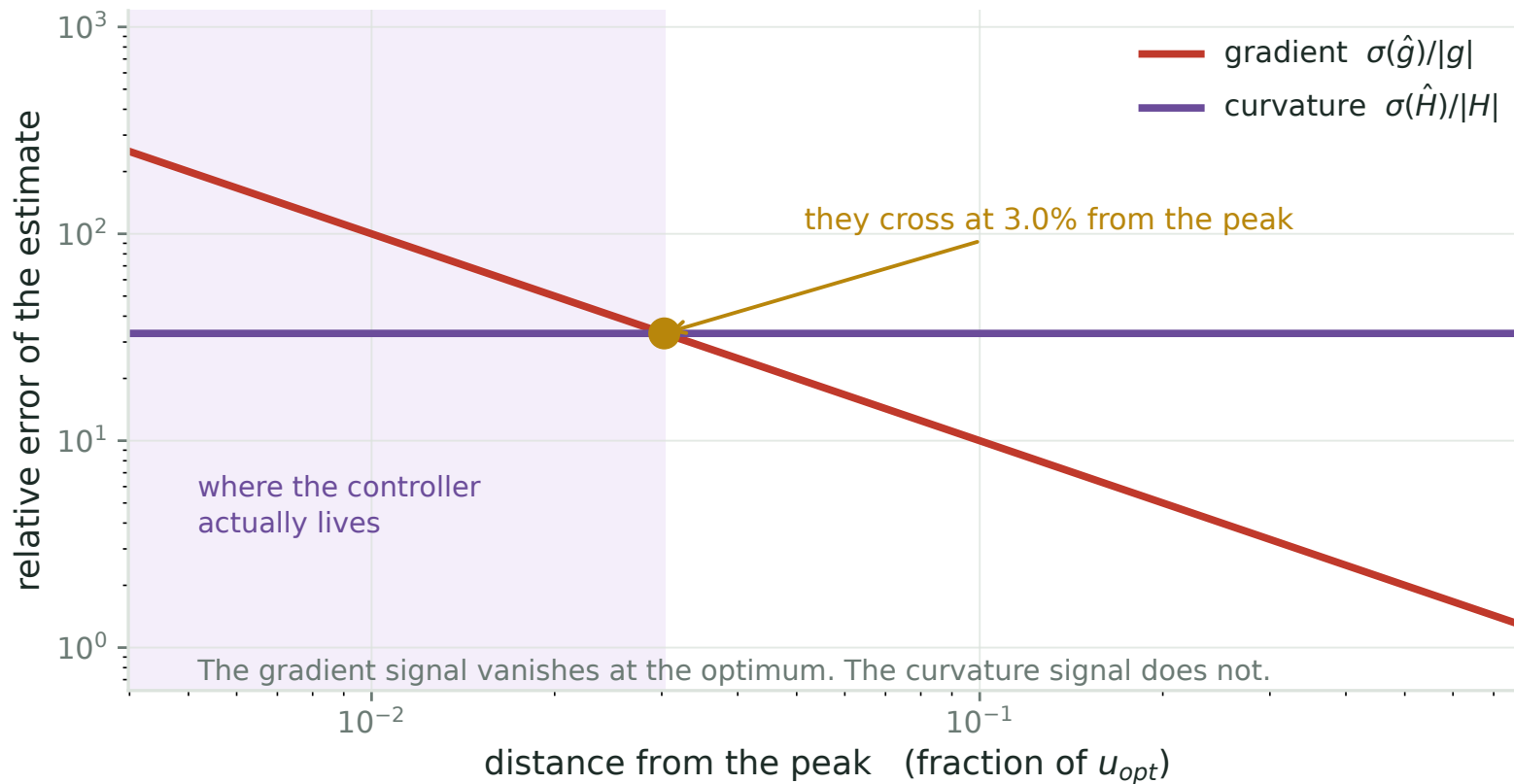
Signal amplitudes are ga at ω and $\frac{1}{4}Ha^2$ at 2ω ; the demodulators divide by a and $a^2/8$. With D the distance from the peak and $g \approx HD$ near it:

$$\frac{\sigma(\hat{H})}{\sigma(\hat{g})} = \frac{8}{a} 2^{-5/6}, \quad \frac{\text{rel}_H}{\text{rel}_g} = \frac{\sigma(\hat{H})/|H|}{\sigma(\hat{g})/|g|} = D \frac{\sigma(\hat{H})}{\sigma(\hat{g})}$$

rel_g **diverges** at the optimum because $g \rightarrow 0$ there. rel_H does not, because H does not. So there is always a neighborhood inside which the curvature is the better conditioned of the two, and it ends at $D = a 2^{5/6}/8 = 3.0\%$ of u_{opt} .

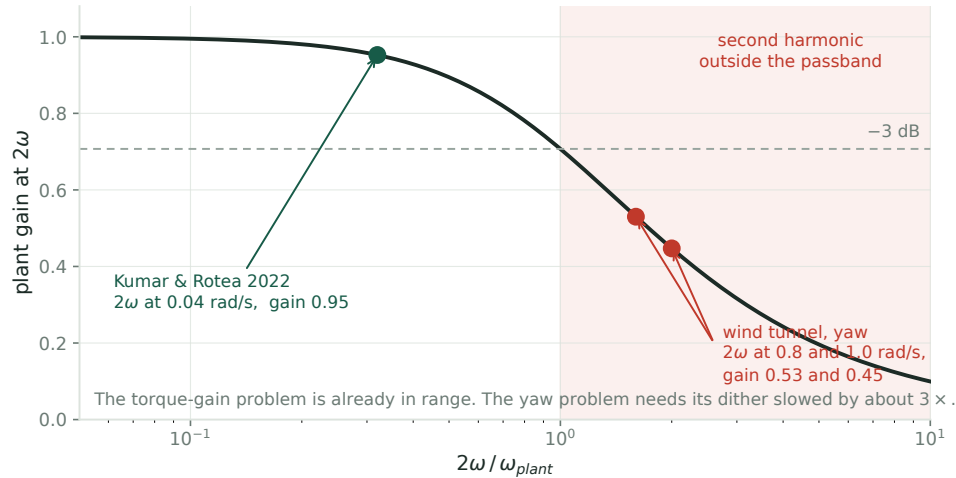
$\sigma(\hat{H})/\sigma(\hat{g})$ carries units of $1/u$, so it is not a bare number: 33 per u_{opt} , 15 per $\text{N}\cdot\text{m}\cdot\text{rpm}^{-2}$. Only D is unit-free.

The crossover, and where the loop sits



After convergence the loop sits about 0.5% out, the finite-difference bias from Q1. That is comfortably inside the crossover. **The regime where Newton methods are supposed to be impossible is the regime the turbine operates in.**

The second harmonic has to fit inside the rotor



The rotor is a low-pass filter. If 2ω falls outside its passband the second harmonic is attenuated and phase-shifted, and the curvature estimate is biased by an amount nobody is tracking.

V	rotor $1/\tau$	$(1/\tau)/2\omega$ at $T = 150$ s	T needed for $2\times$ margin
4 m/s	0.078	0.93	321 s
8 m/s	0.157	1.87	160 s
12 m/s	0.235	2.80	107 s

At 4 m/s the second harmonic is **above** the rotor corner: the Hessian channel is outside the plant. And shortening T to converge faster pushes 2ω further out, so speed and curvature pull in opposite directions.

The timescales are reversed, and that is mostly good

What Zap assumes

$$\varepsilon_n / \alpha_n \rightarrow \infty$$

Jacobian estimate fast,
parameter slow.



What this plant does

θ^* drifts over minutes.
 H drifts over months, with erosion.
The Jacobian is the SLOW variable.

Keep Meyn's recursion; feed it the demodulated $\hat{H}_n$ as innovation:

$$\hat{A}_{n+1} = \hat{A}_n + \gamma(\hat{H}_n - \hat{A}_n), \quad \text{so} \quad N_H = 2/\gamma - 1$$

Small γ makes $\hat{A}$ slow and quiet. Erosion at months against
an estimator at a day is still a two-timescale separation,
only the other way up.

Zap's separation is structurally unavailable here: $\hat{H}$ and θ both update once per dither period. It is also **not needed**, because H barely moves where it matters, only a 20% swing across the terminal scatter.

Sizing the rolling estimate

Do not use the demodulated $\hat{H}_n$ as the gain. Use it as the **innovation** in Meyn's own recursion, with a small γ :

$$\hat{A}_{n+1} = \hat{A}_n + \gamma(\hat{H}_n - \hat{A}_n), \quad \text{Var}(\hat{A}) = \frac{\gamma}{2 - \gamma} \text{Var}(\hat{H}), \quad N_H = \frac{2}{\gamma} - 1$$

One period of $\hat{H}$ is far too noisy to use raw, so the recursion is not an optimization but the only thing that makes the estimate exist. Sizing it from the spectral argument, with $\sigma(\hat{H}) = 7.1 |H|$ per period:

target rel. error	N_H	$1/\gamma$ (periods)	wall clock at $T = 600$ s
50%	201	101	17 h
20%	1255	628	4.4 d

Simulation says this is **optimistic by about $4\times$ in $\sigma(\hat{H})$** . The measured requirement is on the *Simulated* slides below. Quote those, not this table.

Meyn's structure survives intact; only the ordering flips. An estimator running at days against erosion at months is still a two-timescale separation, though the measured margin below is nearer $5\text{--}10\times$ than $100\times$.

It cannot help the transient. Over $u : 4.5 \rightarrow 2.2$, $|H|$ moves $1.8\times$ and not monotonically, dipping to $0.72\times$ near $u = 3$. A day-long estimator cannot track minutes, so early convergence stays uncalibrated.

Zap against averaging

What averaging gives you

The optimal asymptotic covariance. Free. No Jacobian, no second demodulator, no frequency budget, no bandwidth constraint.

What the matrix gain adds

Gain calibration, transient behavior, and conditioning in more than one dimension. It does **not** improve the terminal variance.

Both reach the same asymptotic covariance, the one no unbiased scheme beats:

$$\Sigma^{\text{PR}} = A^{-1} \Sigma_W A^{-\top}, \quad A = \nabla^2 J(\theta^*), \quad \Sigma_W = \text{covariance of the estimator noise}$$

Averaging gets there by keeping a running mean of the iterates, $\bar{\theta}_n = \frac{1}{n} \sum_{k \leq n} \theta_k$, and needs no A at all. Zap gets there by estimating A and inverting it.

So the case for a matrix gain here rests on **tuning effort and conditioning**, not on terminal variance. Argued on variance it loses to a running mean. And note neither is in the published turbine loops, which are constant-gain and so do not average down at all.

Without extremum seeking

Drop the demodulation apparatus and run stochastic approximation directly.

What is left when ESC goes

What gets deleted

The dither frequency, the phase compensation θ , the high-pass and low-pass filters, the lock-in multiplier, the requirement that 2ω fit inside the plant passband.

What is left

$$\theta_{n+1} = \theta_n + \alpha_n G_n \hat{g}_n$$

Perturb, hold, measure, update. The SA recursion with nothing wrapped around it.

An iteration is now a small number of **held operating points**, each averaged long enough to see through the turbulence, rather than a continuous sinusoid the machine rides.

The gradient: SPSA

Draw Δ_n with independent ± 1 entries, perturb **every** parameter at once, and take one difference:

$$\hat{g}_n = \frac{\bar{y}(\theta_n + c\Delta_n) - \bar{y}(\theta_n - c\Delta_n)}{2c} \begin{bmatrix} \Delta_{n1}^{-1} \\ \vdots \\ \Delta_{np}^{-1} \end{bmatrix}$$

where $\bar{y}$ is the mean of $\ln P$ over a held window.

Two measurements, whatever p is. That is the property the deterministic scheme cannot match, and it is the entire case for going stochastic on this plant.

Two replacements for the second harmonic

The second harmonic is gone with the tone. Two replacements:

Spall's 2SPSA

A second perturbation $\tilde{\Delta}_n$, two more gradient estimates, and a symmetrized outer product

$$\hat{H}_n = \frac{1}{2} \left[\frac{\partial \hat{g}_n}{\partial c} \tilde{\Delta}_n^{-\top} + (\cdot)^{\top} \right]$$

Rank two per iteration, four measurements.

Gaussian probe and Stein

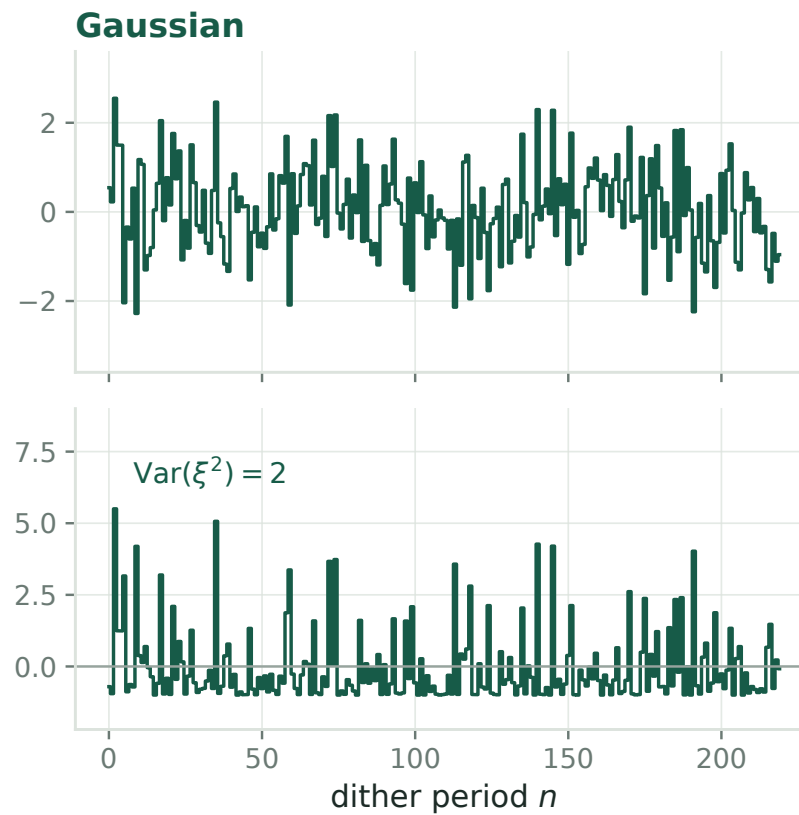
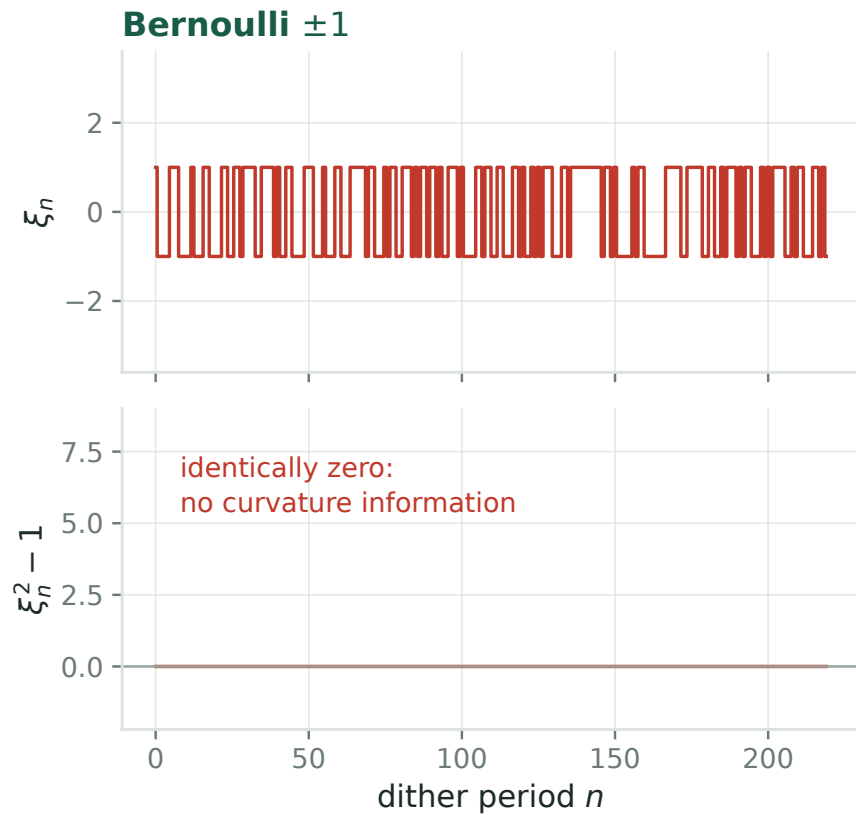
For $\xi \sim N(0, I)$, the second-order Stein identity gives

$$\mathbb{E}[y(\xi\xi^{\top} - I)] = a^2 \nabla^2 \Gamma$$

so $\hat{H}_n = \langle y(\xi\xi^{\top} - I) \rangle / a^2$. No second perturbation.

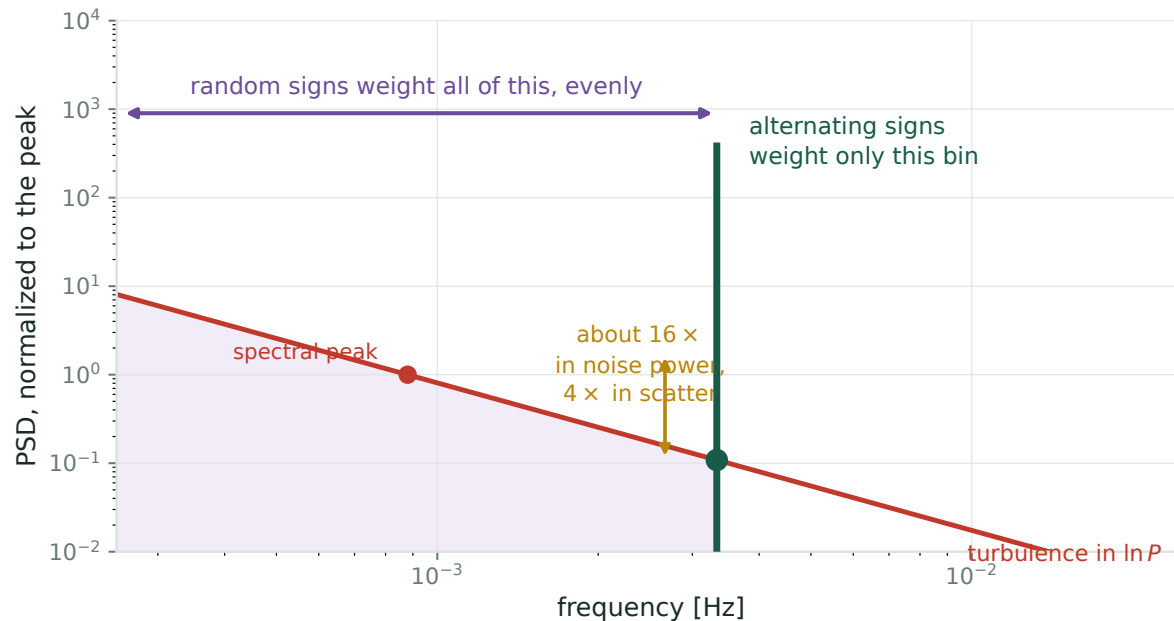
Either way $A_{n+1} = \hat{H}_n$, the running average is Zap's (8.51a), and the regularized inverse is unchanged. **Spall's 2SPSA is Zap with SPSA supplying the Jacobian.**

The Bernoulli trap



The natural stochastic analogue of the [CLR19] square wave is a random ± 1 sign sequence. It gives the gradient, but $\xi^2 \equiv 1$, so $\text{Var}(\xi^2) = 0$ and the Stein estimator **divides by zero**. Bernoulli probing carries no curvature information at all.

Where each scheme weights the turbulence



Both schemes difference two windows, so both reject DC. The difference is *where* each weights the turbulence.

An alternating sign sequence concentrates the estimator's noise weighting in one bin; a random sequence is white and weights everything below its Nyquist evenly. The tone therefore *can* be placed clear of the energy-containing range.

Two problems with pressing this. The ratio depends entirely on how long each random sign is held, which sets how many independent samples a run contains, and it swings from $0.4\times$ at a 28 s hold to $2.3\times$ at a full 150 s hold. And the published tone is not placed clear of anything: Kaimal $fS(f)$ peaks at $U/4L = 0.0059$ Hz and the dither sits at 0.0067 Hz. **It is on the peak.**

Frequency slots against measurements

The deterministic Hessian needs a separate channel for every entry of $\nabla^2 \Gamma$, at a distinct frequency product, none colliding with any ω_k or $2\omega_k$, all inside the plant bandwidth.

parameters p	frequency slots needed	2SPSA measurements
1, torque gain	2	4
2, torque and pitch	6	4
6, yaw, optimized jointly	58	4

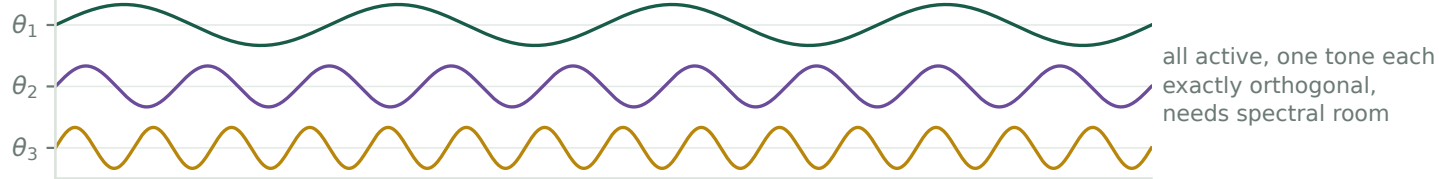
At $p = 1$ the tone wins on every axis. At $p = 6$ the plan needs 58 slots, so the averaging window has to stretch to 49 minutes before noise is even considered, and the stochastic version becomes the only one that runs.

The two Hessians, compared

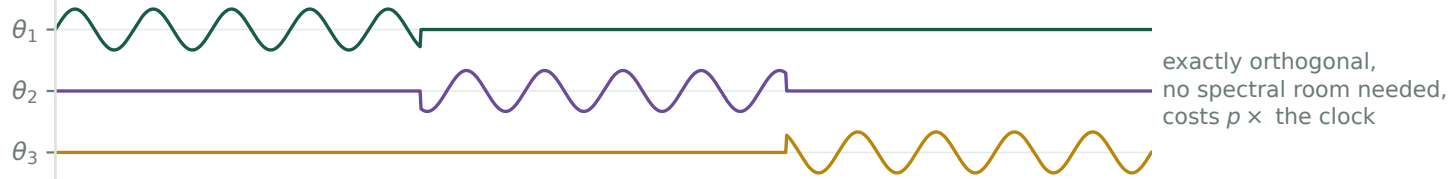
Same matrix, two ways of measuring it, and the difference is combinatorial.

Why frequencies at all

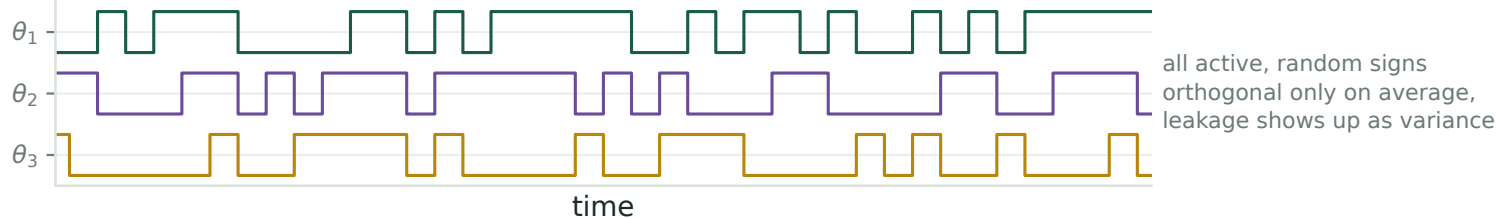
Frequency division: what ESC does



Time division: one parameter at a time



Code division: what SPSA does



There is one measurement, $\ln P(t)$, and p unknowns. The dither frequencies are a **multiplexing scheme**, nothing more. Distinct ω_i make the demodulation channels orthogonal, since $\langle \sin \omega_i t \sin \omega_j t \rangle = \frac{1}{2} \delta_{ij}$.

What the Hessian costs in frequency

The quadratic term produces products, and $\sin \omega_i t \sin \omega_j t = \frac{1}{2} [\cos(\omega_i - \omega_j)t - \cos(\omega_i + \omega_j)t]$. So H_{ij} appears at **sum and difference** frequencies and H_{ii} at $2\omega_i$. Every member of

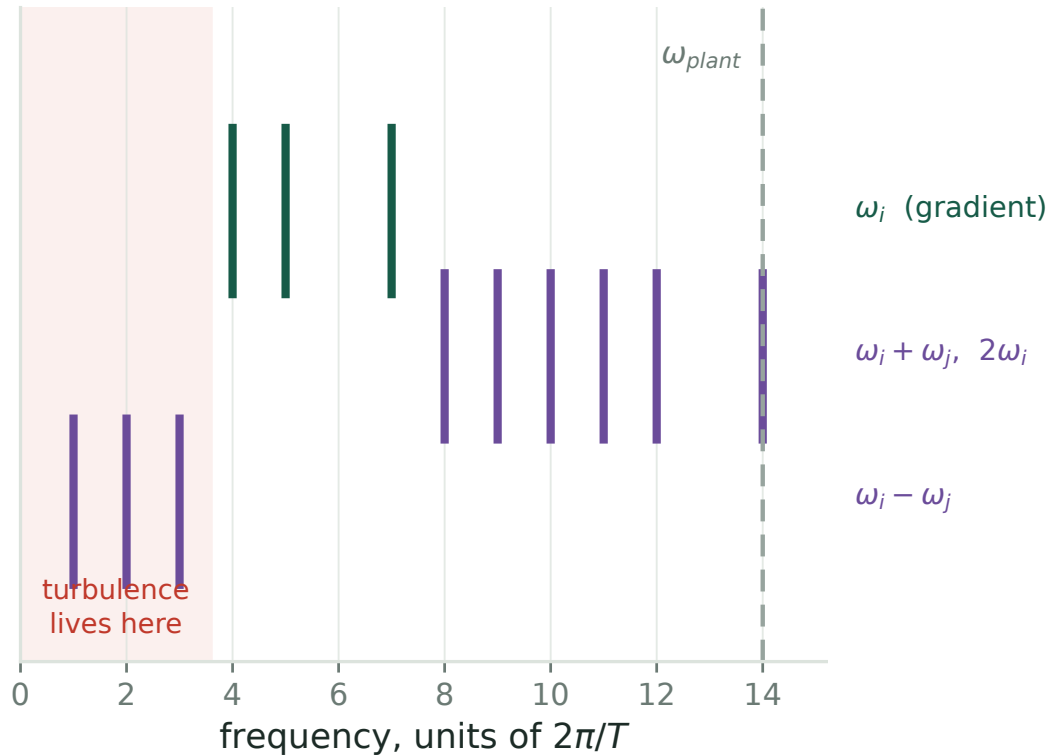
$$\{\omega_k\} \cup \{2\omega_k\} \cup \{\omega_i \pm \omega_j\}$$

must be distinct, and all of it must fit below ω_{plant} .

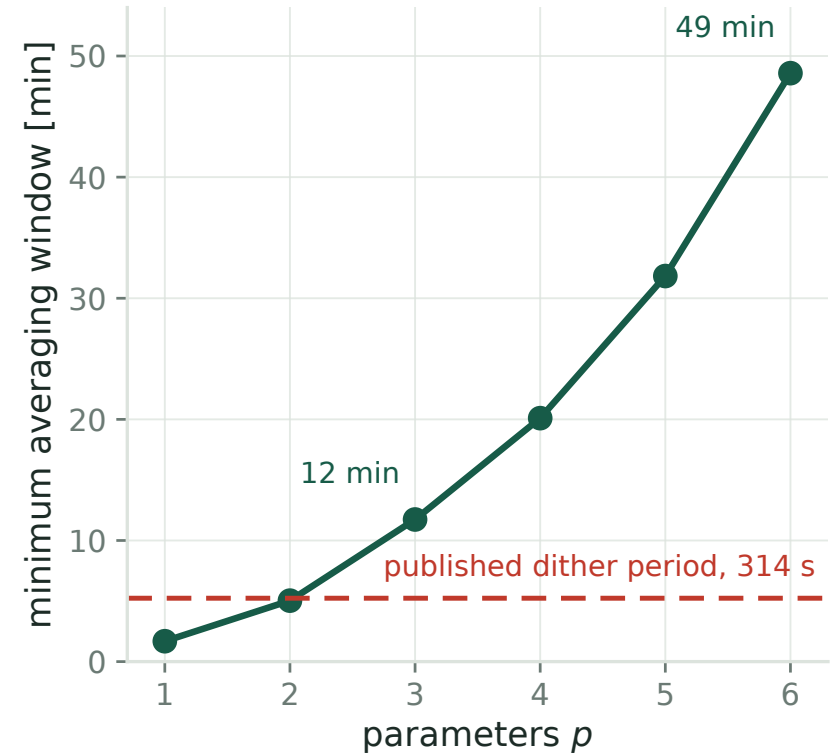
That is stronger than a Sidon set, because the gradient tones must also avoid every second-order product. A plain Golomb ruler is necessary and not sufficient.

And it grows quadratically

$p = 3$: twelve channels, all distinct



what the frequency plan costs



p	a valid tone set	widest product	minimum window
1	{1}	2	101 s
3	{4, 5, 7}	14	704 s
6	{4, 10, 17, 26, 28, 29}	58	2915 s

← Wind Turbine Control · Q4, the Matrix Gain

Head to head

	ESC, second harmonic	SA, 2SPSA or Stein
separation	frequency division	code division
orthogonality	exact, over one period	in expectation only
measurements per update	0 extra, same window	4 extra
spectral demand	grows as p^2	none
noise weighting	one chosen bin	whole band below Nyquist
scatter in $\hat{g}$	reference	0.4 to 2.3 $\times$, hold-dependent
settling dead time	none, rides continuously	about 5τ per held point
perturbation seen by the drivetrain	smooth sinusoid	step changes
degenerate cases	tones colliding with products	Bernoulli probe, $\text{Var}(\xi^2) = 0$

The tone wins on orthogonality, measurement count and actuation smoothness, and loses on spectral demand, which is the row that decides the farm-scale problem. **Noise weighting is a draw**, not a win: it would favor the tone only if the tone were placed off the turbulence peak, and at $T = 150$ s it is not.

Reading the table honestly

$$p \leq 2$$

The frequency plan is trivial and the tone is better on every axis that matters. There is no argument for random probing here.

$$p \geq 4$$

The plan needs a 20 to 50 minute window before noise even enters. Random probing is not merely competitive, it is the only scheme that runs.

The load row deserves more weight than it usually gets. Kumar and Rotea 2022, Figure 13 of [RKAJ24]: the **single** step in torque gain when the controller switches on raises drivetrain torsion DEL by 70% at 4 m/s. SPSA makes a step like that every iteration, in both directions, and nobody has costed that out.

The synthesis this points to

Both columns assume the Hessian must be measured at the same rate as the gradient. Nothing in the problem requires that.

Gradient

Keep the tone. Continuous, gentle, low-noise, and only p frequencies are needed when no products have to be resolved.

Curvature

A deliberate experiment, run rarely. H drifts with erosion, over months. Once a week is often enough.

In between

Use the last estimate as a slowly refreshed preconditioner, which is Zap with the timescales put the right way round.

Run rarely, the curvature experiment can be **time-division multiplexed**: sweep one parameter at a time. No frequency plan, no 2ω constraint, no p^2 bandwidth, and no random probing either.

Simulated

Everything so far is algebra on a fitted curve. This section runs it.

How the simulation is built

The plant

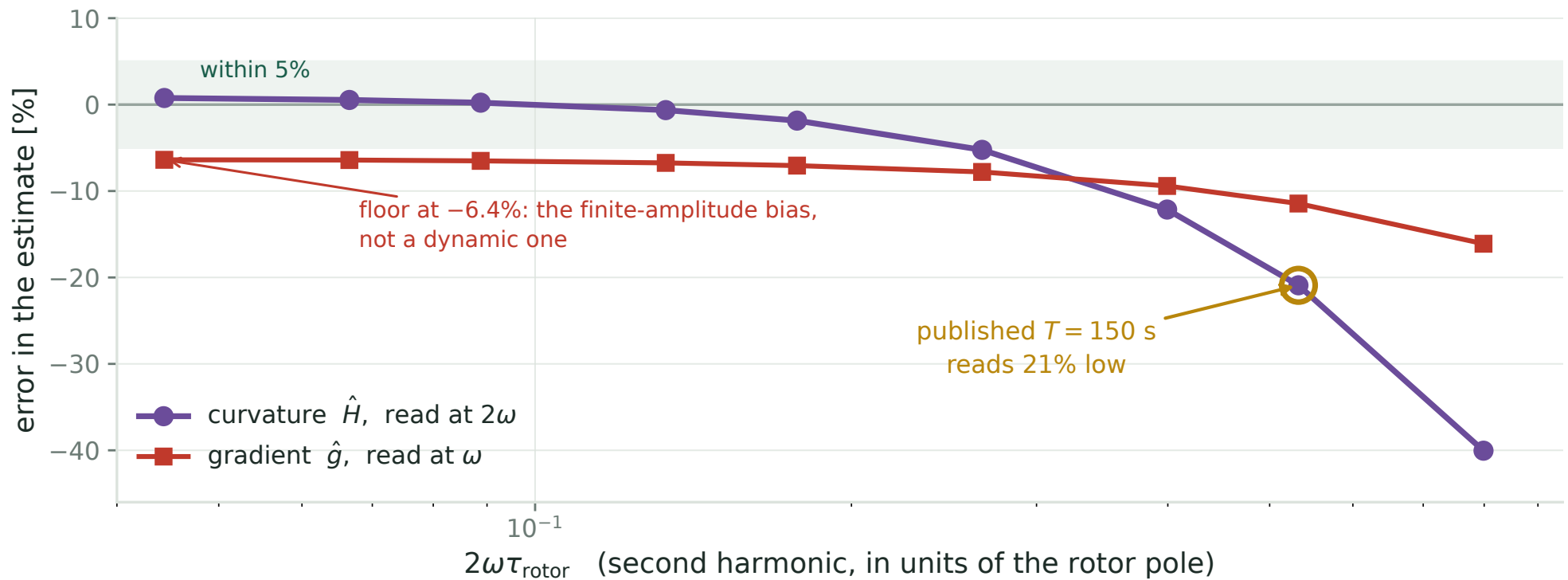
NREL 5 MW rotor. $I_{\text{eff}}\dot{\Omega} = \tau_{\text{aero}}(V, \Omega) - k\Omega^2$ integrated with **diffirax** (Tsit5, adaptive, rtol 10^{-6}). Heier C_P remapped to $(\lambda^*, C_P^{\text{max}}) = (7.5, 0.49)$.

The loop

Sinusoidal dither $u_n + a \sin \omega t$ at $a = 13.6\%$ of u^* , the [KR22] amplitude. Demodulate $\ln P$ at ω and 2ω over each period, then update. Turbulence is an IEC Kaimal realization at TI = 10%.

One knob, **sharpness**, scales the deviation of C_P from its peak. It leaves λ^* and C_P^{max} untouched and multiplies $|H|$ at the optimum by exactly that factor, verified to three digits. That is how the plant curvature is varied without changing anything else.

Simulated: the demodulator, and what the rotor does to it



Gain held 5% off the peak in steady wind, so the true J'' is known and the only error is the estimator's. $\hat{H}$ is **exact to 0.8%** once $2\omega\tau_{\text{rotor}} \lesssim 0.1$, so the identity survives the rotor dynamics. At the published $T = 150$ s, $\hat{H} = 0.79 J''$: right sign, right shape, magnitude a fifth too small.

Why it reads low, and how to undo it

Fitting the whole sweep, the bias is $\hat{H}/J'' = (1 + (2\omega\tau)^2)^{-1} = |G(2\omega)|^2$, to 1.2 percentage points rms. **Two** factors of $|G|$, for two separate reasons:

attenuation

The rotor is a low-pass between torque gain and power, so the second harmonic arrives smaller by $|G(2\omega)| = (1 + (2\omega\tau)^2)^{-1/2}$.

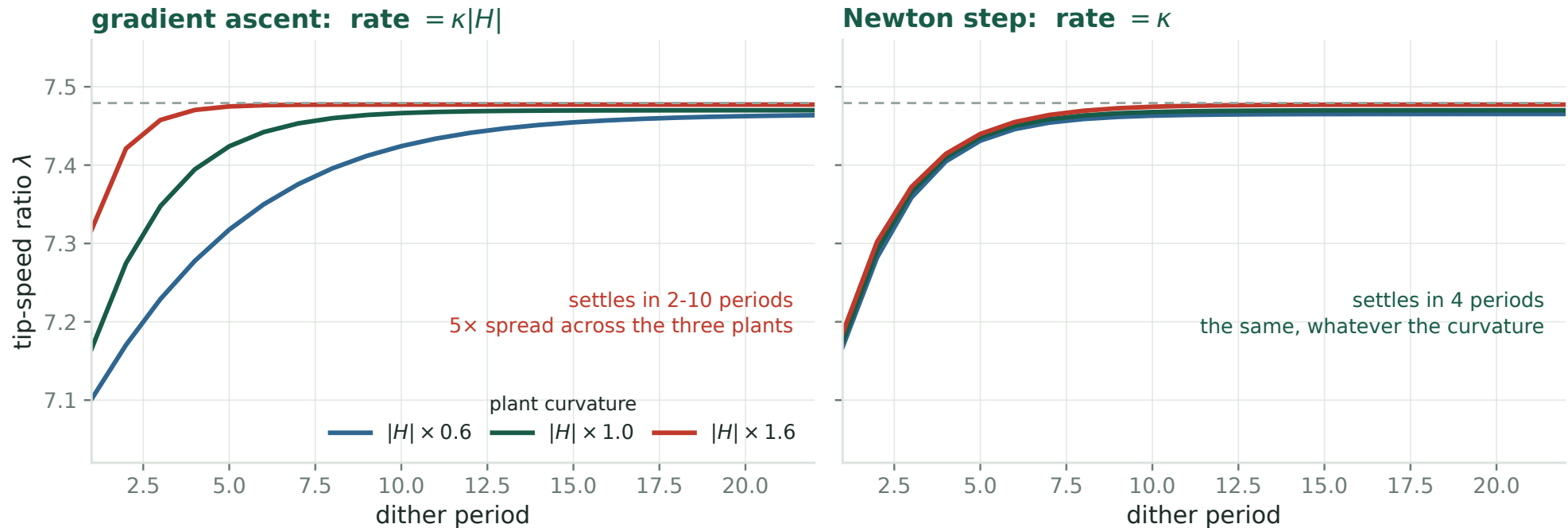
phase mismatch

It also arrives late, by $\phi = -\arctan 2\omega\tau$. Correlating against an in-phase $\cos 2\omega t$ recovers $\cos \phi$, which is the same factor again.

The gradient is attenuated too, just less, so the two **partly cancel** in $\hat{g}/\hat{H}$. Dynamic parts only: $\hat{g}/g = 0.946$, $\hat{H}/H = 0.791$, so the Newton step is inflated by $1.20\times$, matching the predicted $|G(\omega)|^2/|G(2\omega)|^2 = 1.199$.

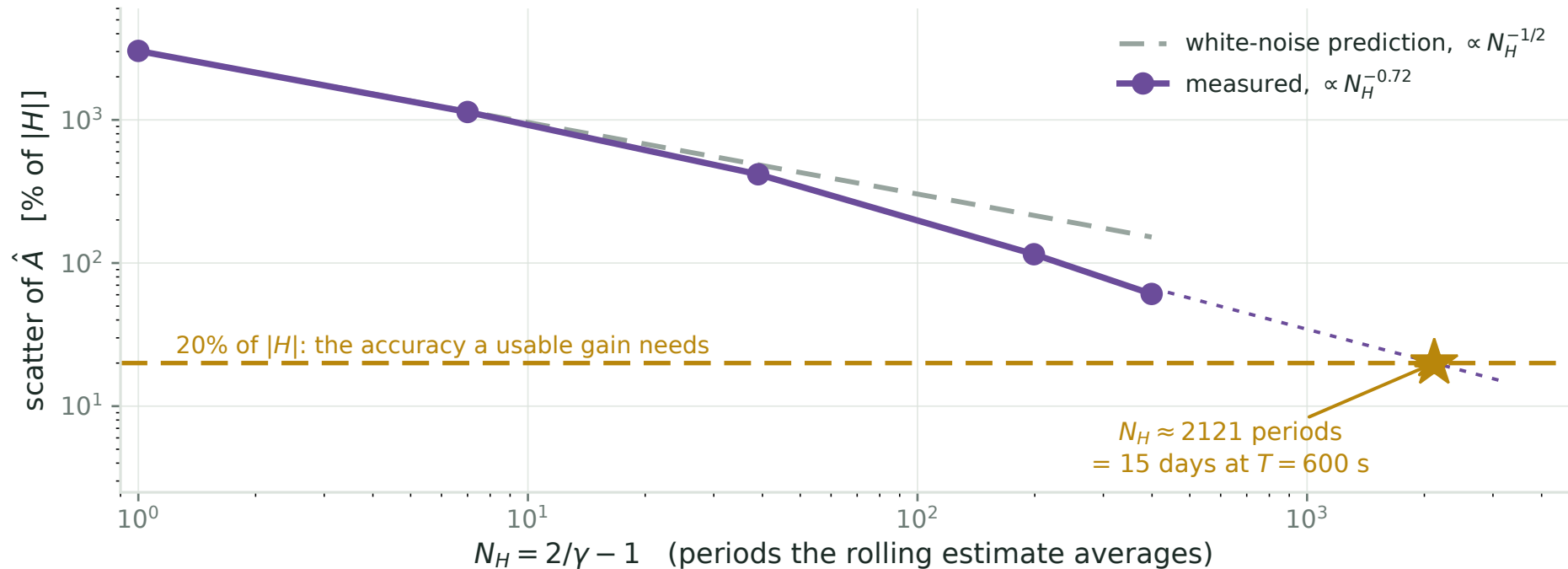
So **divide it out**: $\hat{H}_{\text{corr}} = \hat{H} (1 + (2\omega\tau)^2)$, with τ from the rotor model you already have. A calibration, not a stability threat: at the published $K = 0.4$ the inflated gain is 0.48 against a bound of 2. But left uncorrected it defeats the point of a matrix gain, which was to make κ be the settling time.

Simulated: the curvature does leave the loop gain



κ tuned **once**, for $K = 0.4$ on the $|H| \times 1.0$ plant, never retuned; all six runs start at $1.2 u^*$ in steady wind. Gradient ascent settles in 10, 5 and 2 periods as $|H|$ goes $0.6 \rightarrow 1.0 \rightarrow 1.6$, a 5× **spread**. The Newton step settles in 4 on all three, a 1.0× **spread**. The central claim of the deck, and it holds.

Simulated: what the gain estimate costs



Gain held at u^* , open loop, 300 periods of Kaimal turbulence. One period's $\hat{H}$ scatters by 3000% of $|H|$, four times the spectral estimate. Useless on its own.

The rolling filter rescues it **faster than white noise would**, $N_H^{-0.72}$ against $N_H^{-1/2}$, because consecutive errors are negatively correlated in band-limited turbulence. Reaching 20% needs $N_H \approx 2100$ periods, about 15 days at $T = 600$ s.

Verdict

Tone, feasible now

Single-turbine torque gain, $p = 1$, so no frequency plan to solve. But **slow the dither**: simulation puts $2\omega\tau = 0.53$ at $T = 150$ s, where $\hat{H}$ reads 21% low. $T = 600$ s brings that under 1%.

Tone, after retuning

Two parameters, (u, β) . Three demodulation channels to place, and the bandwidth is there for them.

Random probing

Yaw across a farm, optimized jointly. This is not what the tunnel experiment does: it runs six scalar loops on clusters, which is how it keeps $p = 1$.

Both versions share everything downstream of the Jacobian: the averaging recursion, the regularized inverse, the Polyak-Ruppert tail. Only the way A_{n+1} is measured changes, so the two are the same program with two front ends.

What I would test first

P0, two weeks

Run the existing LP-ESC at a fixed torque gain, demodulate at 2ω , and compare $\hat{H}$ against the curvature of the simulator's own C_P surface. Pure measurement, no loop closed.

Fitting 2ω inside the rotor

Does $\hat{H}$ converge to within, say, 20% over a plausible averaging window, and does its scatter match the $(8/a)2^{-5/6}$ prediction?

If $\hat{H}$ can be measured open-loop, everything else in this deck is engineering. If it cannot, the matrix gain is dead and Polyak-Ruppert averaging is the whole answer to Rotea's question.

A matrix gain for extremum seeking

The Jacobian Zap needs is measurable: one octave up from the gradient, or in the second moment of a random probe.

The mapping

f is a gradient estimate, from a tone or from a random perturbation. A is the curvature of $\ln C_P$, read at 2ω or from a second moment. $\hat{A}$ is the Hessian that erosion moves.

The open part

Which front end to use is set by p , not by taste. Still open: a convergence theory for the reversed timescale ordering this plant actually wants.

Meyn, *Control Systems and Reinforcement Learning*, §8.5 · Ghaffari, Krstic & Nesic, *Automatica* 2012 · Spall, *IEEE TAC* 45 (2000) 1839–1853 · Kumar & Rotea, *Energies* 2022 · Rotea, Kumar, Aju & Jin, *J. Phys. Conf. Ser.* 2767 (2024) 032043

References

[R17] M. A. Rotea, *Logarithmic power feedback for extremum seeking control of wind turbines*, IFAC-PapersOnLine **50**(1) (2017) 4504–4509. doi:10.1016/j.ifacol.2017.08.381

[CLR19] U. Ciri, S. Leonardi & M. A. Rotea, *Evaluation of log-of-power extremum seeking control for wind turbines using large eddy simulations*, Wind Energy **22** (2019) 992–1002. doi:10.1002/we.2336

[KR22] D. Kumar & M. A. Rotea, *Wind turbine power maximization using log-power proportional-integral extremum seeking*, Energies **15** (2022) 1004. doi:10.3390/en15031004

[KR24] D. Kumar & M. A. Rotea, *Optimal tip-speed ratio for degraded blades*, Wind Energy Science **9** (2024) 2133–2146. doi:10.5194/wes-9-2133-2024

[RKAJ24] M. A. Rotea, D. Kumar, E. J. Aju & Y. Jin, *Multi-row extremum seeking for wind farm power maximization*, J. Phys. Conf. Ser. **2767** (2024) 032043. doi:10.1088/1742-6596/2767/3/032043

[MGR24] S. P. Mulders, N. Gallo & M. A. Rotea, *Analysis of extremum seeking control for wind turbine torque controller optimization by aerodynamic and generator power objectives*, ACC (2024). arXiv:2407.08059

[GKN12] A. Ghaffari, M. Krstić & D. Nešić, *Multivariable Newton-based extremum seeking*, Automatica **48** (2012) 1759–1767. doi:10.1016/j.automatica.2012.05.059

[S00] J. C. Spall, *Adaptive stochastic approximation by the simultaneous perturbation method*, IEEE Trans. Automat. Contr. **45** (2000) 1839–1853. PDF

[LM23] C. Lauand & S. Meyn, *Quasi-stochastic approximation: design principles with applications to extremum seeking control*, IEEE Control Systems Magazine **43** (2023). doi:10.1109/MCS.2023.3291884

[A22] N. J. Abbas et al., *A reference open-source controller for fixed and floating offshore wind turbines*, Wind Energy Science **7** (2022) 53–73. doi:10.5194/wes-7-53-2022

[J09] J. Jonkman, S. Butterfield, W. Musial & G. Scott, *Definition of a 5-MW reference wind turbine*, NREL/TP-500-38060 (2009). PDF · [IEC] IEC 61400-1 ed. 3, normal turbulence model.